Energy Harbor Nuclear Corp
Perry Nuclear Power Plant
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February 22, 2023
L-23-015

10 CFR 50.73(a)(2)(iv)(A)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2023-001, "Manual ECCS Actuation Following Automatic Reactor Trip." There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Glendon Burnham, Manager – Regulatory Compliance, at (440) 280-7538.

Sincerely,

Rod L. Penfield

Enclosure:
LER 2023-001

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III Regional Administrator

NRC FORM 366 (01-10-2023) | U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/

1. Facility Name		2. Docket Number	3. Page
Perry Nuclear Power Plant	☒ 050 / ☐ 052	00440	1 OF 3

4. Title: Manual ECCS Actuation Following Automatic Reactor Trip

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
01	05	2023	2023	- 001 -	00	02	XX	2023	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	098

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or in NRC Form 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
George Dujanovic, Staff Nuclear Engineering Specialist, Regulatory Compliance	440-280-5200

13. Complete One Line for Each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	SJ	CPU	F180	Y	X	SJ	JX	X999	Y

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On January 5, 2023, at 1242, with the reactor in Mode 1 and at 98% rated thermal power, an automatic reactor trip occurred due to lowering water level in the reactor vessel. The High Pressure Core Spray (HPCS) and Reactor Core Isolation Cooling (RCIC) systems were manually initiated to restore and control water level in the reactor vessel.

The direct cause of the reactor trip was attributed to the failure of the Digital Feedwater Control System (DFWCS) to control reactor water level, following corrective maintenance on one of the power sources to the system.

A latent configuration control issue in the system caused unexpected indication losses during the maintenance. Upon re-energizing the power source, two component failures in DFWCS, and an overloaded communications network, caused the system to lower reactor water level, which resulted in a reactor trip.

The failed components were replaced, and the reactor was re-started on January 8, 2023.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) due to an automatic Reactor Protection System (RPS) actuation and the manual initiation of two Emergency Core Cooling System (ECCS) systems.

1. FACILITY NAME	2. DOCKET NUMBER		3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant	☒ 050 / ☐ 052	00440	2023	- 001	- 00

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

EVENT DESCRIPTION

On January 5, 2023, planned corrective maintenance was being performed on a voltage-regulating transformer [XFMR] which is one of the redundant power sources to the Digital Feedwater Control System (DFWCS) [SJ]. When the transformer was de-energized for the work to be performed, Operators received some unexpected losses of DFWCS indication, resulting in the loss of all of the DFWCS Operator interface screens. Reactor power, pressure and level remained stable at this time, indicating that DFWCS maintained control.

Based on the plant remaining stable, a decision was made to continue with the work on the transformer.

Upon the post-maintenance energization of the voltage regulating transformer, a reactor water level transient of three inches occurred, and then stabilized. A walkdown of DFWCS identified a failed level control processor module [CPU] and a degraded power supply [JX]. The decision was made to perform a Nodebus Check (online diagnostic test) to restore the lost indication in the system. Due to the loss of all the DFWCS interface screens, the Operators did not recognize that the system demanded zero level to the DFWCS controllers during a re-sync of the level control processor modules. This zero-demand signal was sent to the Reactor Feed Pump Turbines (RFPTs) [P] governor [65] thereby causing the RFPTs to reduce output to idle and minimum flow. The reduced feedwater flow resulted in the reactor water level lowering to Level 3, which caused an automatic reactor trip at 1242. Operators then manually initiated two Emergency Core Cooling Systems (ECCS), High Pressure Core Spray (HPCS) [BG] and Reactor Core Isolation Cooling (RCIC) [BN] to restore and control reactor water level. The Motor Feed Pump became available at 1251 and was started at 1253 to control reactor water level. At 1254, the Motor Feed Pump was controlling reactor water level through DFWCS.

The Condition Report investigation revealed that the cause of the reactor trip was attributed to the failure of a DFWCS level control processor module along with a degraded power supply, concurrent with a DFWCS network communications overload.

Additionally, a configuration control issue was found with some of the DFWCS media translators [CNV] being connected to the incorrect power source. This resulted in the unexpected loss of additional DFWCS workstations and indications to the Operators when power was removed for the maintenance. It also caused increased DFWCS network communication traffic, due to the loss of interface screens hindering the Operator's ability to acknowledge alarms.

NRC notification (EN 56298) was made on January 5, 2023, at 1539, under 10 CFR 50.72(b)(2)(iv)(A) - ECCS Injection and 10 CFR 50.72(b)(2)(iv)(B) - RPS Actuation.

The faulty DFWCS power supply and the failed secondary control processor module were replaced. The configuration control issue was also corrected with the media translators. Post maintenance testing for DFWCS was performed satisfactorily, and a successful reactor startup was completed on January 8, 2023.

CAUSES OF EVENT

The direct cause of the automatic reactor trip was attributed to the failure of one of the DFWCS level control processor modules, along with an intermittent power supply failure, and combined with a network traffic overload which overall resulted in a zero percent output demand signal to the Reactor Feed Pump Turbine (RFPTs) governor, causing the RFPTs to reduce speed to idle and minimum flow output.

NARRATIVE

CAUSES OF EVENT (continued)

Other cause identified:

A latent human performance issue occurred during an upgrade of the media translators, and a configuration control issue occurred with correct power sources to the media translators. This went undiscovered until this event occurred.

EVENT ANALYSIS

A Probabilistic Risk Assessment (PRA) bounding evaluation was performed for the January 5, 2023 event. An analysis of this uncomplicated plant scram results in a delta CDF and a delta LERF that are well below the acceptable thresholds of 1.0E-06/yr and 1.0E-07/yr, respectively, as discussed in Regulatory Guide 1.174. The risk of this event is therefore considered to be of very low safety significance in accordance with the Regulatory Guidance.

CORRECTIVE ACTION

The DFWCS defective power supply and the secondary control processor module were replaced. Configuration control issue with the media translators was corrected.

PREVIOUS SIMILAR EVENTS

There have been no Licensee Event Reports in the past three years related to DFWCS.